

Effective Strip Noise Removal for Low-Textured Infrared Images Based on 1-D Guided Filtering

Yanpeng Cao, Michael Ying Yang, and Christel-Loic Tisse

Abstract—Infrared images typically contain obvious strip noise. It is a challenging task to eliminate such noise without blurring fine image details in low-textured infrared images. In this paper, we introduce an effective single-image-based algorithm to accurately remove strip-type noise present in infrared images without causing blurring effects. First, a 1-D row guided filter is applied to perform edge-preserving image smoothing in the horizontal direction. The extracted high-frequency image part contains both strip noise and a significant amount of image details. Through a thermal calibration experiment, we discover that a local linear relationship exists between infrared data and strip noise of pixels within a column. Based on the derived strip noise behavioral model, strip noise components are accurately decomposed from the extracted high-frequency signals by applying a 1-D column guided filter. Finally, the estimated noise terms are subtracted from the raw infrared images to remove strips without blurring image details. The performance of the proposed technique is thoroughly investigated and is compared with the state-of-the-art 1-D and 2-D denoising algorithms using captured infrared images.

Index Terms—Focal plane array (FPA), guided filtering, infrared imaging, nonuniformity correction (NUC), strip noise removal.

I. INTRODUCTION

INFRARED imaging technology provides an important tool in a wide range of military and civilian applications, including thermal imaging, night vision, video surveillance, astronomy, fire detection, and robotics. However, the performance of an infrared imaging system is strongly affected by spatial nonuniformity of its focal plane array (FPA) detectors. Such nonuniformities of FPA detectors will cause obvious and time-dependent fixed pattern noise (FPN), which significantly degrades the quality of its captured infrared images [1], [2]. Since it is difficult to fix these problems on the hardware level, our research motivation is to develop effective image processing algorithms to compensate for hardware defects of infrared detectors.

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In uncooled long-wave infrared (LWIR) imaging systems, nonlinear behaviors of infrared detectors and lack of column cross calibration generate obvious strip noise in captured infrared images [3], [4]. Human observers are particularly sensitive to the resulting high-frequency image artifacts. This kind of complex FPN often makes an infrared image unrecognizable, as shown in Fig. 1(a), and is difficult to remove using traditional calibration-based [5] or scene-based [6]–[9] nonuniformity correction (NUC) techniques.

In the field of computer vision, many effective algorithms have been proposed for image denoising [10]–[13]. However, most 2-D denoising methods will inevitably remove fine details of input images, which is particularly a problem for low-textured infrared images [14]. As shown in Fig. 1(b), it becomes difficult to distinguish targets with insignificant temperature variations (e.g., pedestrians and road signs) in the smoothed output. A number of papers have been published on techniques for removing 1-D stripe noise in infrared images [15]–[17]. These methods typically apply a threshold parameter, which distinguishes image edges and stripe noise, to avoid edge blurring. However, this strategy will falsely remove weak image edges and keep strong stripe noise. The approach in [18] groups pixels that belong to the same readout channel (e.g., within a column) and applies an offset correction model to normalize outputs of these channels so that pixels from adjacent columns produce similar outputs. However, the global model-based solutions [4], [18], [19] do not produce satisfactory destriping results. A significant amount of strips remains visible in their processed results. It is because locally varied strip noise cannot be effectively compensated for using column-wise fixed correction factors. Moreover, most commercial infrared cameras apply temporal image smoothing to suppress strip noise, while such processing will cause image blurring when applied to a dynamic scene (e.g., contains moving objects). Recently, Tendero *et al.* [3], [20], Tendero and Gilles [21], and Delon [22] have proposed effective stripe removal methods based on column-wise midway histogram equalization (MHE). These methods can effectively eliminate stripe noise without blurring fine image details. However, MHE algorithms work by adjusting intensity distribution of pixels within a column, and therefore cannot completely remove strip noise and will often generate undesired image artifacts as shown in Fig. 1(c). Also MHE needs to perform column-based histogram and inverse histogram computations that are not suitable for real-time hardware implementation.

The motivation of our research work is to develop a single-image-based processing algorithm, which is suitable

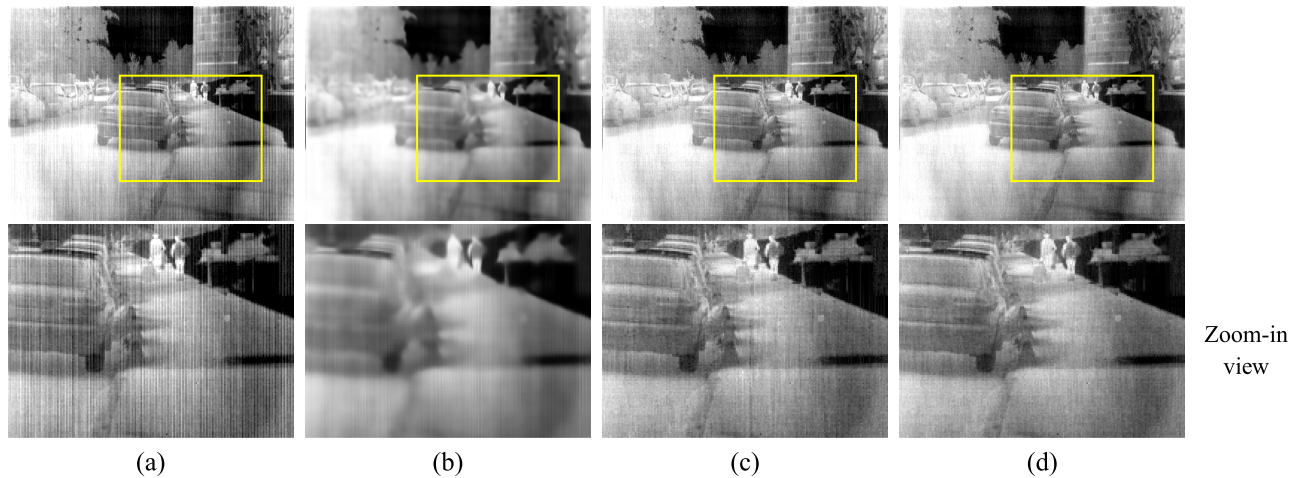


Fig. 1. Strip noise removal of an infrared image. (The image is available under the Creative Commons Attribution (CC-BY) license [20].) (a) Raw image. (b) Denoising result using Non-local means (NL-means) algorithm [11]. (c) Denoising result based on column-wise MHE [20]. (d) Denoising result of our proposed algorithm. Our proposed method can effectively remove significant strip noise but without blurring important image details or causing undesired artifacts. See Section IV for more experimental results.

for hardware real-time implementation, to effectively compensate for strip-type spatial nonuniformity present in infrared images. Our method starts by applying a 1-D row guided filter to perform horizontal edge-preserving image smoothing to remove vertical strips. In this step, the raw infrared data are used as input and guidance images of the 1-D row guided filter. However, the edge-preserving guided filter will still oversmooth low-textured infrared images and the removed noise part contains valuable image details. Therefore, an additional processing step is required to retrieve the high-frequency components corresponding to image texture and structures [23]. Through a thermal calibration experiment, we discover that a local linear model can be applied to approximate the relationship between infrared image data and strip noise for sensors within a column. Based on this novel local linear noise behavioral model, strip noise can be accurately separated from high-frequency signals caused by image texture. Such a decomposition problem is efficiently solved using a modified 1-D column guided filter. The column guided filter takes the extracted high-frequency image part (both strip noise and texture) as input and computes the strip noise term by considering infrared responses of pixels within a column. Only the estimated strip noise is subtracted from the raw infrared image, while the fine details are well preserved to avoid undesired image blurring.

Our proposed method has three major advantages over other state-of-the-art denoising solutions [11]–[13], [16], [18]–[21]. First, this single-image-based NUC method works on static images. Therefore, it does not require processing a number of video frames and will not cause ghosting artifacts. Second, it delivers a better strip noise correction performance compared with the state-of-the-art 1-D and 2-D denoising techniques (see Fig. 1 for comparison). The proposed method can effectively remove significant strip noise without blurring important image details or causing undesired artifacts. As a consequence, visually insignificant but important targets (e.g., pedestrians, vehicles, and road signs) can be clearly identified after the strip removal step. Finally,

our proposed method is based on a sequence of 1-D box filters that can be efficiently implemented in an Field-programmable gate array (FPGA) processor. Moreover, the method is memory efficient. It requires storing only a small part of the raw image in memory to start processing, while other strip removal solutions [16], [18], [19], [21] require storing an entire image frame. Therefore, our proposed destriping method can be easily implemented in an FPGA as an effective preprocessing step to improve the quality of images captured by an infrared camera.

The remainder of this paper is organized as follows. In Section II, we start by reviewing some existing NUC and image denoising solutions. The details of the proposed algorithm are presented in Section III. The novel algorithm is evaluated and compared with the existing state-of-the-art 1-D and 2-D denoising solutions using captured infrared images in Section IV. Finally, conclusions are given in Section V.

II. RELATED WORK

Spatial nonuniformity is an undesirable hardware defect present in infrared FPA detectors. The problem is that different response outputs are returned by individual sensors, even though they receive the same amount of radiant energy and they are fabricated using the same material and technique [24]. Moreover, the characteristics of spatial nonuniformity are not constant and are significantly affected by temperature fluctuations of FPA, optic lens, and other mechanical components [25], [26]. Therefore, the correction factors are temperature dependent and require to be updated frequently.

The most basic and effective NUC solution is based on radiometric calibration. For instance, the two-point calibration method employs two temperature-uniform blackbodies at different temperatures to calculate the gain and bias parameters of a linear NUC model [5]. However, this method will interrupt camera's normal image capturing for a few seconds and is not suitable for real-time infrared imaging applications.

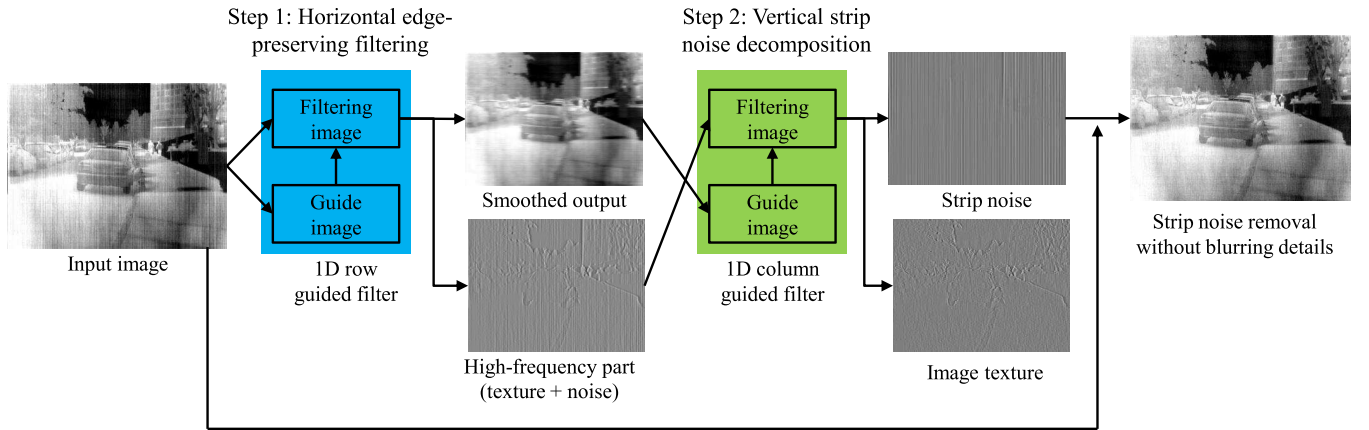


Fig. 2. Processing workflow of the proposed method for strip NUC. This single-frame-based method can accurately remove strip noise without blurring details in low-textured infrared images.

To overcome the limitation, many shutterless methods have been proposed for temperature-dependent NUC through image sequence processing [6], [7], [9], [27]. These scene-based NUC methods rely on either image statistic analysis [6], [27] or temporal motion tracking [7]. The problem of scene-based NUC algorithms is that they usually need to process a number of image frames to estimate stable NUC factors, which is too slow for frame-based correction. Also, if the image sequence does not contain enough scene motions, objects in previous images may appear as the accumulated ghosting artifacts in the current image frame.

Infrared images are typically corrupted by obvious high-frequency strip noise (both spatially and temporally). NUC techniques developed for slowly drifting spatial nonuniformity [6], [9], [27] are not suitable to remove the resulting FPN. In the field of computer vision, a large number of single-frame-based denoising methods have been proposed [10]–[12], [28], [29]. These denoising methods typically separate an input image into a smooth part and a high-frequency part. Ideally, the smoothed output looks like the original image and the removed high-frequency part contains as little structure as possible. However, these general denoising methods cannot completely separate image texture from stripe noise, and therefore will inevitably remove fine details of infrared images. This is a particularly difficult problem for processing infrared images that contain insufficient amount of texture information [14]. Important targets will no longer be detected/recognized in the oversmoothed output images.

A number of papers have been published on techniques for removing 1-D stripe noise [15]–[17]. These methods typically apply a threshold parameter that distinguishes image edges and stripe noise to avoid image blurring. However, this strategy will falsely indicate that image edges with amplitudes less than the threshold are treated as stripe noise and removed. Stripe noise with an amplitude greater than the threshold is regarded as an image edge and retained. The approach in [18] groups pixels from the same readout channel (e.g., within a column) and applies a linear correction model to normalize the outputs of these channels so that adjacent channels produce pixels with the same mean and standard deviation. Zhao *et al.* [17] proposed a single-frame scene-based NUC

algorithm using gradient-based regularization. The solution aims to seek the optimal image with a vertical gradient as close to the original image as possible and makes the energy of the horizontal gradient as small as possible. This method will also blur images with vertical features. Tendero *et al.* [20] and Tendero and Gilles [21] proposed improved stripe removal methods based on column-wise MHE. They assume that the difference between two adjacent columns is statistically small and further apply the MHE operation to equalize their corresponding histograms. This method does not require separating image edges and stripe noise, and therefore can effectively eliminate stripe noise without blurring edges. To the best of our knowledge, MHE algorithm is one of the top performing strip noise removal solutions for infrared images. However, the algorithm needs to perform column-based histogram equalization and inverse histogram computation, which are slow for real-time hardware implementation. As well, the MHE algorithm is not capable of removing locally significant strip noise and sometimes generates undesired image artifacts.

III. PROPOSED APPROACH

Our proposed infrared image strip noise removal algorithm consists of two consecutive processing steps: 1) performing horizontal edge-preserving image smoothing and 2) decomposing strip noise from the overall extracted horizontal high-frequency signals. Only the computed noise components are subtracted from the raw input to avoid undesired image blurring effects. A modified 1-D guided filter [30] is applied for both processing steps. The complete processing pipeline is schematically shown in Fig. 2.

A. Guided Image Filtering [10], [31]

A guided filter smooths an input image p under the guidance of another image I . The local linear model of the guided filter assumes that filtered output q is a linear transform of the guidance I in a window ω_k centered at pixel k

$$q_i = a_k I_i + b_k \quad \forall i \in \omega_k \quad (1)$$

where (a_k, b_k) are some linear coefficients assumed to be constant in window ω_k . A square window is used in the

original formula of 2-D guided filtering. In our method, a 1-D row window and a 1-D column window are used for horizontal edge-preserving image filtering and vertical strip noise decomposition, respectively. Then guided filter seeks for coefficients (a_k, b_k) that minimize the difference between the output q and the input p within ω_k as follows:

$$E(a_k, b_k) = \sum_{i \in \omega_k} ((a_k I_i + b_k - p_i)^2 + \epsilon a_k^2) \quad (2)$$

where ϵ is the regularization term penalizing large a_k . It is noted that parameter ϵ controls the strength of local averaging and its effect is similar to the range variance parameter σ_r^2 in a bilateral filter [12]. Minimizing (2) is a least-square problem within ω_k , and its solution is given by

$$a_k = \frac{\frac{1}{|\omega|} \sum_{i \in \omega_k} I_i p_i - \mu_k \bar{p}_k}{\sigma_k^2 + \epsilon} = \frac{\frac{1}{|\omega|} \sum_{i \in \omega_k} I_i p_i - \mu_k \bar{p}_k}{\frac{1}{|\omega|} \sum_{i \in \omega_k} I_i^2 - \mu_k^2 + \epsilon} \quad (3)$$

$$b_k = \bar{p}_k - a_k \mu_k \quad (4)$$

where μ_k and σ_k^2 are the mean and variance of guidance image I in window ω_k , $\bar{p}_k$ is the mean of input image p in ω_k , and $|\omega|$ is the number of pixels in ω_k . Since a pixel i is involved in a number of overlapping windows, the final filtering output is given by

$$q_i = \frac{1}{|\omega|} \sum_{k|i \in \omega_k} (a_k I_i + b_k) = \bar{a}_i I_i + \bar{b}_i \quad (5)$$

where $k|i \in \omega_k$ defines a number of overlapping box windows ω_k covering the pixel i , $\bar{a}_i = (1/|\omega|) \sum_{k|i \in \omega_k} a_k$, and $\bar{b}_i = (1/|\omega|) \sum_{k|i \in \omega_k} b_k$ are the coefficient averages of all windows overlapping pixel i . The guided filter is mostly used as an efficient alternative to the popular edge-preserving smoothing operator—bilateral filter [12]. It has been proven that the kernel weights of the guided filter can be explicitly expressed by

$$W_{i,j}(I) = \frac{1}{|\omega|^2} \sum_{k|(i,j) \in \omega_k} \left(1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon} \right). \quad (6)$$

Therefore, the weight term is large for pixel pairs on the same side of the edge and small otherwise. It means that inputs p_i and p_j of pixel i and j are averaged only if their corresponding guidance images I_i and I_j are similar (on the same side of a structure). Instead of directly performing slow translation-variant and range-variant convolutions, the guided filter efficiently makes use of a number of box filters to compute the filtered output based on (3)–(5).

More generally, the guided filter is not a simple smoothing filter. Due to the local linear model $q = aI + b$, output q is locally a scaling (plus an offset) of the guidance image I . This property makes it possible to transfer the structure from guidance I to output q . The guided filter has been applied in many interesting filtering applications like dehazing [10], saliency detection [32], image colorization [33], illumination rendering [34], and cost volume filtering [35]. In this paper, we apply a modified 1-D guided filter for horizontal edge-preserving image filtering (Section III-B) and vertical strip noise decomposition (Section III-C).

B. Horizontal Edge-Preserving Filtering

The infrared image model is generally formulated as follows:

$$v_i = u_i + s_i + t_i \quad (7)$$

where v_i is the raw infrared image, u_i is its smoothed version, s_i is the strip noise, and t_i is the image texture and other forms of image noise. Our objective is to eliminate strip noise without losing image texture. Given a strip noise corrupted infrared image, our method first applies a 1-D row guided filter to remove high-frequency signals in the horizontal direction. The modified 1-D row guided filter takes the raw infrared data v_i as its input and guidance images, and smooths similar pixels within a row window (size $1 \times w$). In our implementation, we set $w = 9$ for a 320×240 resolution infrared image.

After the horizontal edge-preserving filtering step, the raw infrared image is divided into a smoothed image part and some horizontal high-frequency components as

$$v_i = u_i + n_i \quad (8)$$

where n_i contains strip noise, image texture, and some other image noise. The key idea of edge-preserving filtering is using a parameter (ϵ for the guided filter and σ^2 for the bilateral filter) to distinguish high-variance edges and low-variance noise. Only noise is smoothed, and image edges are retained to avoid image blurring. However, this strategy will fail when working on low-textured infrared images since strip noise is more significant than many image details. Fig. 3 shows the outputs of 1-D row guided filter with various values of parameter ϵ . If we set ϵ to a low value ($\epsilon = 0.1^2$), many obvious vertical strips still remain in the smoothed output. On the other hand, setting ϵ to a large value ($\epsilon = 0.4^2$) can completely remove vertical strips, but will oversmooth fine details in low-textured infrared images. As shown in Fig. 3 (top), it becomes difficult to distinguish targets with small temperature variations (e.g., trees and pedestrians) in the filtered output. As shown in Fig. 3 (bottom), the smoothed-out high-frequency part contains a significant amount of detail information. In our proposed method, we set a high ϵ ($\epsilon = 0.4^2$) here to completely remove strip noise and involve a consecutive step to recover the high-frequency components corresponding to image texture and edges. This step is a standard application of guided filter for edge-preserving denoising (input image is equal to guidance image), while the smoothing is performed only in the horizontal direction and no edges in the vertical direction get blurred. Other structure-preserving filters (e.g., bilateral filter [12] or rolling guidance filter [36]) can also be modified for this step. As shown in Fig. 4, the strip removal results using a 1-D bilateral filter and a 1-D rolling guidance filter are similar to the result of the 1-D guided filter. The advantage of guided filter is that it can be efficiently implemented using a sequence of box filters [31].

C. Vertical Strip Noise Decomposition

Recently, numerous research works have attempted to improve the performance of image denoising algorithms

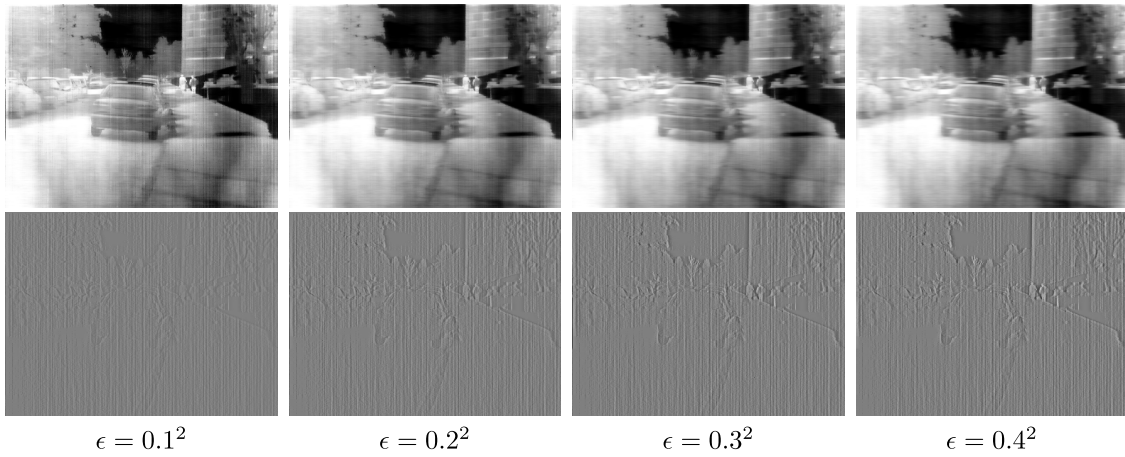


Fig. 3. Strip noise removal of Fig. 1(a) using the 1-D row guided filter with various values of parameter ϵ . Top: the smoothed output. Bottom: the extracted horizontal high-frequency components. The images are normalized to the 0–1 value range by an affine contrast change for visualization.



Fig. 4. Destriping results of Fig. 1(a) using a 1-D bilateral filter (left) and a 1-D rolling guidance filter (right). The range variance is set at 0.3 and the smoothing window width is set at 9 for both 1-D bilateral and 1-D rolling guidance filters. The iteration number of rolling guidance is set at 4. The images are normalized to the 0–1 value range by an affine contrast change for visualization.

through utilizing prior knowledge on the characteristics of natural images [37]. The key idea is that image denoising can be performed not only using the image itself but rather using a suitable prior on noise statistics. In this paper, we set up the following thermal calibration experiment to investigate the characteristics of strip noise in infrared images so as to develop better infrared image destriping solutions.

First, we make use of a ceramic infrared bulb to heat up a black metal plate (blackbody) unevenly. Then, we use a commercial uncooled LWIR thermal imaging camera with $17\text{-}\mu\text{m}$ pixel size to capture 640×480 resolution infrared images of the blackbody. In this step, we manually set off its on-chip strip noise cancelation algorithm. As shown in Fig. 5, the captured infrared image contains a smoothly varying intensity field that is caused by uneven heat distribution of the blackbody, some obvious vertical strip noise, but no major texture or edges. The shape of the intensity field is adjusted by changing the position of the heater behind the blackbody.

The purpose of such an experimental setup is twofold. First, it enables us to generate different responses at individual sensor detectors. As shown in Fig. 5, the infrared image of an unevenly heated object smoothly changes from dim to bright for pixels within a single column. The second advantage of this setup is that we can easily apply a smoothing operator (e.g., averaging filter or Gaussian filter) to divide a captured infrared image into a smooth part and a high-frequency part. Since the blackbody contains no texture, the smoothed image

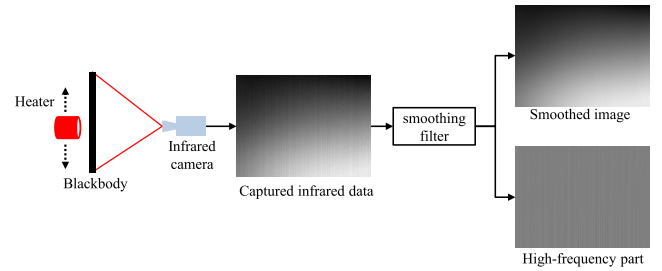


Fig. 5. Thermal calibration experiment for strip noise characterization. Given an infrared image of an unevenly heated blackbody, we apply a smoothing filter (e.g., averaging or Gaussian) to divide the image into a smooth part and a high-frequency part. Since the blackbody contains no texture or edges, the smoothed image is the noise-free infrared data and the extracted high-frequency components are mostly strip noise. Note that the images are normalized to the 0–1 value range by an affine contrast change for visualization.

is the noise-free infrared data and the extracted high-frequency components are mainly caused by strip noise. Through this experimental setup, we can systematically examine the characteristics of strip noise and learn its relationship to pixel received infrared responses.

In Fig. 6, we plot infrared data u_i and strip noise s_i of pixels at a number of selected columns (columns 100–600). It is observed that a smooth (locally linear) curve can be applied to characterize their relationship. Based on the experiment result, a local linear model, similar to (1), is used to approximate the formulation of strip noise for pixels at a single column as follows:

$$s_i = a_{c_k} u_i + b_{c_k} \quad \forall i \in \omega_{c_k} \quad (9)$$

where the linear coefficients (a_{c_k}, b_{c_k}) are constant in a local column window ω_{c_k} . The derived noise model described in (9) satisfies the key assumption of the guided filter that a local linear relationship exists between the guidance image and the filtering output within a defined window. Therefore, we can make use of a 1-D column guided filter to further extract strip noise s_i from high-frequency signals n_i by considering the smoothed image u_i as

$$s_i = \bar{a}'_i u_i + \bar{b}'_i. \quad (10)$$

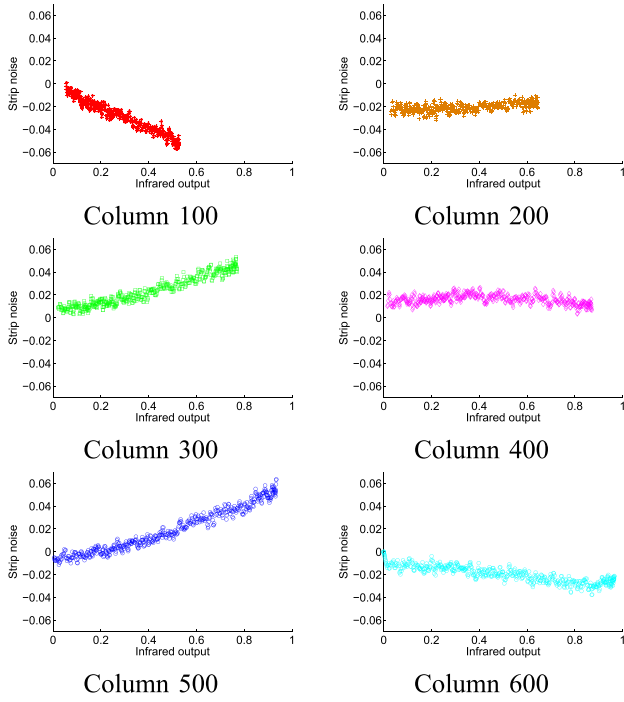


Fig. 6. Infrared data and strip noise of pixels at a number of selected columns (columns 100–600). The infrared data are normalized to the 0–1 value range by an affine contrast change.

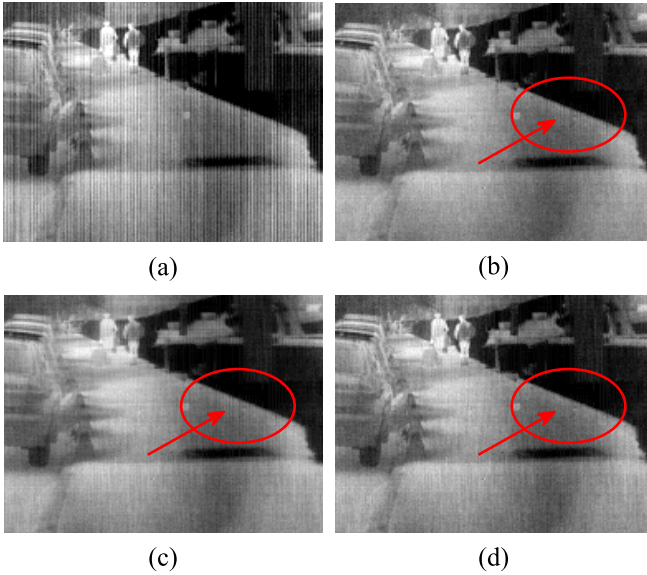


Fig. 7. Example of strip noise removal for a captured infrared image. (a) Zoomed-in patch. (b) Result of 1-D local averaging. (c) Result of applying the standard 1-D column guided filter—the guidance image is n itself. (d) Result of our proposed 1-D column guided filter. It is observed that our proposed 1-D column guided filter can better remove strip noise in the boundary area between the dark and bright regions. Please zoom in to see the details in ellipses.

Here, n_i and u_i are used as input and guidance images to compute $\bar{a}'$ and $\bar{b}'$ based on (3)–(5). Based on (6), the 1-D column guided filter averages n_i and n_j within a local column window ω_{c_k} if their corresponding guidance images u_i and u_j are similar. For a column window that crosses a boundary, as shown in Fig. 7, strip noise is computed for the dark and bright regions separately. Therefore, our

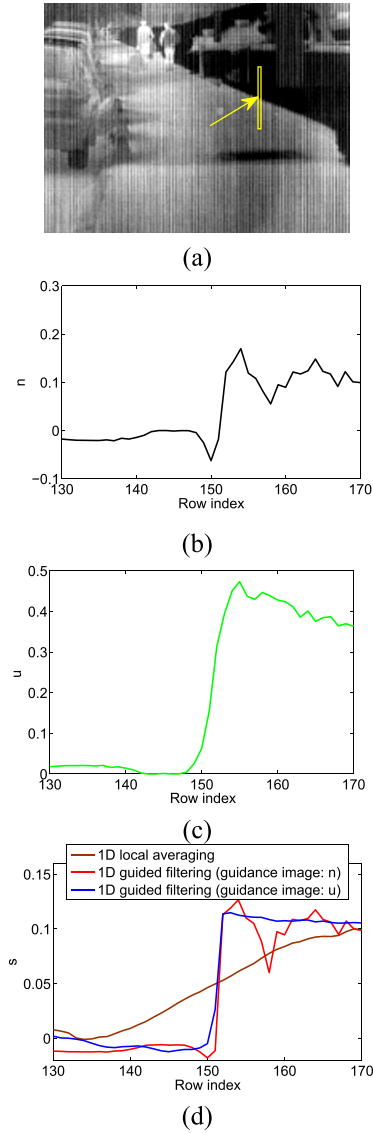


Fig. 8. (a) Signal processing results for the highlighted column window. (b) Extracted high-frequency signal n . (c) Smoothed signal u . (d) Computed strip noise s using different filtering methods. In our implementation, we set $\epsilon = 0.2^2$ for the 1-D column guided filter.

proposed 1-D column guided filter can better estimate strip noise in the boundary area between dark and bright regions (see Fig. 7 for comparison).

We need to point out that this step is not equal to applying a standard 1-D column guided filter to smooth n_i (using n_i for both input and guidance images). The reason we use u_i instead of n_i as the guidance image is that our thermal calibration experiment observes a local linear relationship existing between s_i and n_i and such a relationship (9) satisfies the key assumption of guided filtering. As well, the computed signal n_i is highly dynamic as shown in Fig. 8(b) and cannot provide stable guidance to do local filtering. Figs. 7 and 8 show a comparison of results obtained using different filtering techniques.

An important parameter for accurate vertical strip noise decomposition is the 1-D column window height h . It defines how generally we assume the linear model $s_i = a_{c_k}u_i + b_{c_k}$ holds valid. Fig. 9 shows the decomposition results using the

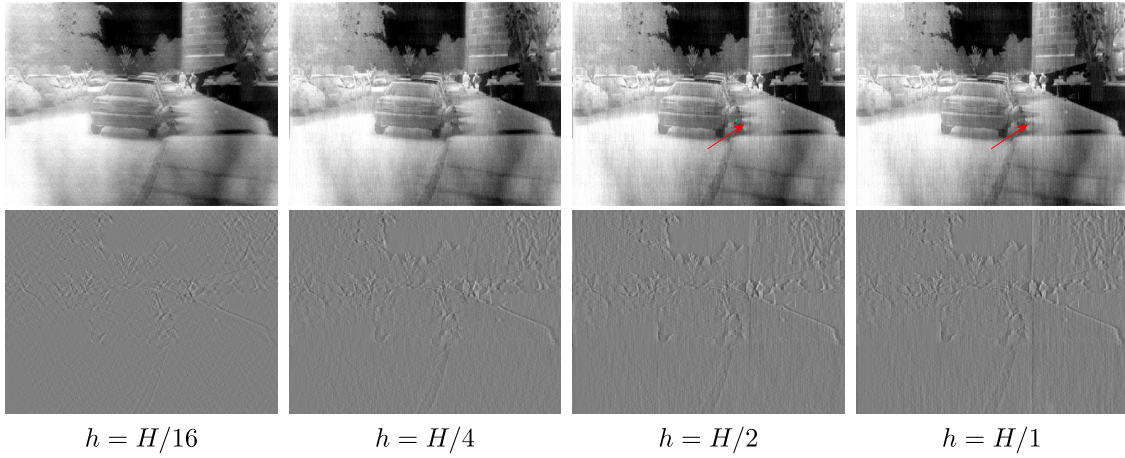


Fig. 9. Top: the strip removal results using the 1-D column guided filter of various heights h . Bottom: the recovered image texture. If we set h to a small value ($H/16$), strip noise cannot be well separated from image texture, and thus the output image is blurred. If we set h to a large value ($H/2$ or H), local strip noise cannot be accurately removed using a global linear model. In our implementation, we set $h = (H/4)$.

1-D column guided filter with various heights. First, h should not be set to a small value since linear coefficients (a_{c_k}, b_{c_k}) cannot be robustly estimated based on the data of a limited number of pixels. Strip noise will not be well separated from image texture, and the output image still gets blurred as shown in Fig. 9. As well, h should not be set to a large value. For instance, if we set h to the image height H , it is equal to fitting a linear model between u_i and s_i for the entire image column. As shown in Fig. 9, local strip noise cannot be accurately removed when fixing (a_{c_k}, b_{c_k}) for the full column ($h = H$). A global linear model holds valid only for columns without significant response variations (e.g., columns 100 and 200 in Fig. 6). For columns covering a wide temperature range (e.g., columns 500 and 600 in Fig. 6), a local linear correction model generates more accurate noise removal results. In our implementation, we set $h = (H/4)$.

D. FPGA Implementation

We briefly discuss the complexity of the proposed method for FPGA hardware implementation. Our method consists of two consecutive processing steps: 1) performing horizontal edge-preserving image smoothing and 2) decomposing strip noise from the overall extracted horizontal high-frequency signals. Both steps make use of a modified 1-D guided filter. It is noted that the average terms shown in (3)–(5) can be efficiently implemented in an FPGA using a sequence of box filters [30], [38]. As shown in Fig. 10, the 1-D guided filter circuit consists of six 1-D box filters and associated combinational arithmetic circuits, which calculate formulas (3)–(5). The 1-D box filter can be efficiently implemented in an FPGA based on a fast calculation algorithm [39]. A register keeps the current summed value of the filter kernel $\text{Sum}(t)$ and the oldest pixel value P_{t-} . When the kernel reads the data of the next pixel P_{t+} , a new summed value $\text{Sum}(t+1)$ is simply calculated as $\text{Sum}(t+1) = \text{Sum}(t) - P_{t-} + P_{t+}$.

Moreover, the proposed method is memory efficient. For a raster scanned infrared image, our proposed method requires buffering only part of a raw image to start processing.

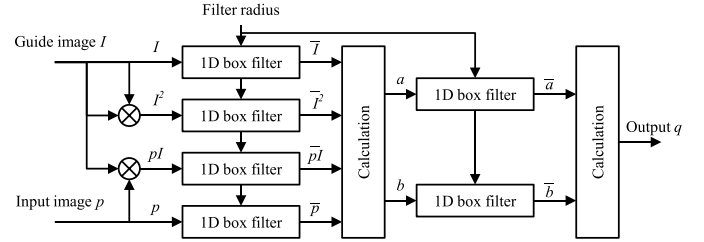


Fig. 10. Block diagram of the 1-D guided filter FPGA implementation.

After receiving $h = (H/4)$ rows of the raw image data, our algorithm can compute the denoised output (the horizontal edge-preserving filtering requires one row of data and the vertical strip noise decomposition requires h rows of data, respectively). In comparison, other strip removal solutions [16], [21] need to compute statistics for a full column of pixels, and thus require storing an entire image frame to start computing the corrected output for pixels on the first row. This advantage makes the proposed method suitable for real-time FPGA implementation.

IV. EXPERIMENTAL RESULTS

In this section, the performance of the proposed technique is thoroughly investigated and further compared with the state-of-the-art 1-D and 2-D image denoising algorithms using captured infrared images. We make use of some publicly available infrared images under the CC-BY license [20] and some of our own captured infrared images to demonstrate that our method is applicable to images captured by different types of infrared devices.

A. Comparison With 2-D Image Denoising Algorithms

First, we apply some state-of-the-art 2-D edge-preserving filtering algorithms to infrared images corrupted by obvious strip noise [20]. Specifically, we consider Block Matching 3D (BM3D) filtering [13] and NL-means filtering [11]. We set the smoothing parameters $\gamma = 0.3$ for both BM3D filtering

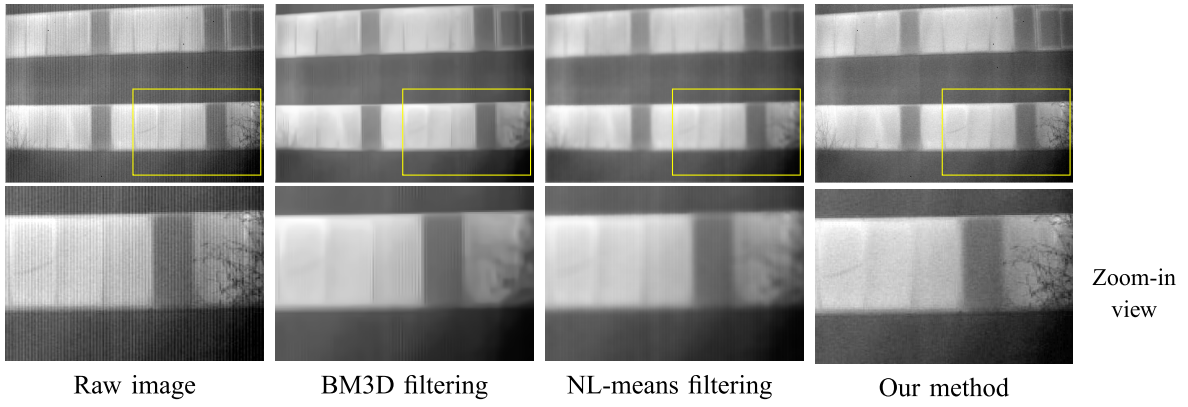


Fig. 11. Top: the raw image and strip removal results using state-of-the-art 2-D edge-preserving filters and our proposed method. Bottom: zoomed-in views of the highlighted region. Note that visually insignificant but important targets (e.g., trees and leaves) can still be easily identified in our result, while they are missed in the outputs of general 2-D denoising techniques. The input image is available under the CC-BY license [20].

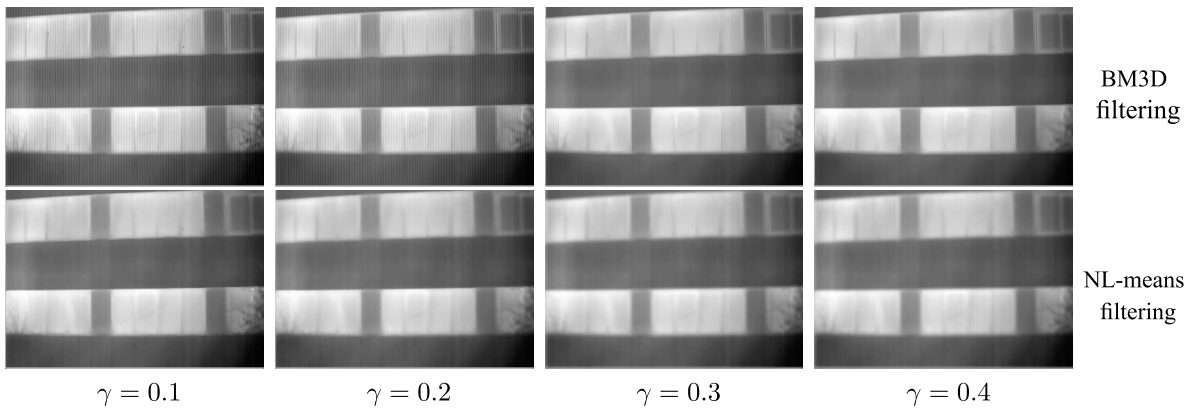


Fig. 12. Denoising results of BM3D filter [13] and NL-means filter [11] using different smoothing parameters.

and NL-means filtering. The search window radius and similarity window radius are set at 5 and 3 for NL-means filtering. For our own method, we set $\epsilon = 0.4^2$ and $w = 9$ for the horizontal edge-preserving filtering step and $\epsilon = 0.2^2$ and $h = H/4$ for the vertical strip noise decomposition step. The comparative results are shown in Fig. 11. It is observed that the state-of-the-art 2-D denoising techniques cannot completely remove strip noise. Their denoising results still contain many obvious vertical strips. Moreover, 2-D denoising methods will inevitably remove fine details of input images and significantly decrease the thermal sensitivity of infrared images. In Fig. 12, we show the denoising results of the BM3D and NLM methods using different smoothing parameters. In low-textured infrared images, sometimes local strip noise is more significant than image details. If we set the smoothing parameter γ to a low value (e.g., $\gamma = 0.1$), many obvious vertical strips still remain visible in the smoothed output. On the other hand, setting the smoothing parameter to a large value (e.g., $\gamma = 0.4$) can better remove vertical strips, but will oversmooth important details in infrared images. The experimental results conclude that the general 2-D edge-preserving denoising algorithms might work well for texture-rich visible images; however, they are not suitable to process low-textured infrared images and will lose many valuable details. Therefore, our research motivation is to

develop a denoising algorithm that can effectively remove strip noise without blurring fine image details. Visually insignificant but important targets (e.g., trees and leaves) can still be easily identified in our filtered output.

B. Comparison With 1-D Image Denoising Algorithms

In this section, we compare our proposed method with a number of state-of-the-art strip correction algorithms. First of all, we choose the MHE method [20] as the baseline method. Its source code is available under the CC-BY license [20]. Based on the assumption that histograms of two adjacent columns are similar, MHE performs horizontal column histogram smoothing within a local window to correct intensity variances caused by strip noise. This method can effectively eliminate stripe noise without blurring edges. To the best of our knowledge, the MHE algorithm is one of the best performing strip noise correction solutions. We also consider two other model-based strip noise compensation solutions [4], [18] in our comparison. The C-model method [4] applies a 1-D smoothing filter (in our implementation, we make use of a 1×9 Gaussian filter) to extract high-frequency signals in the horizontal direction, and then computes a constant term (offset correction model) to add to each column so that the difference between pixels from two

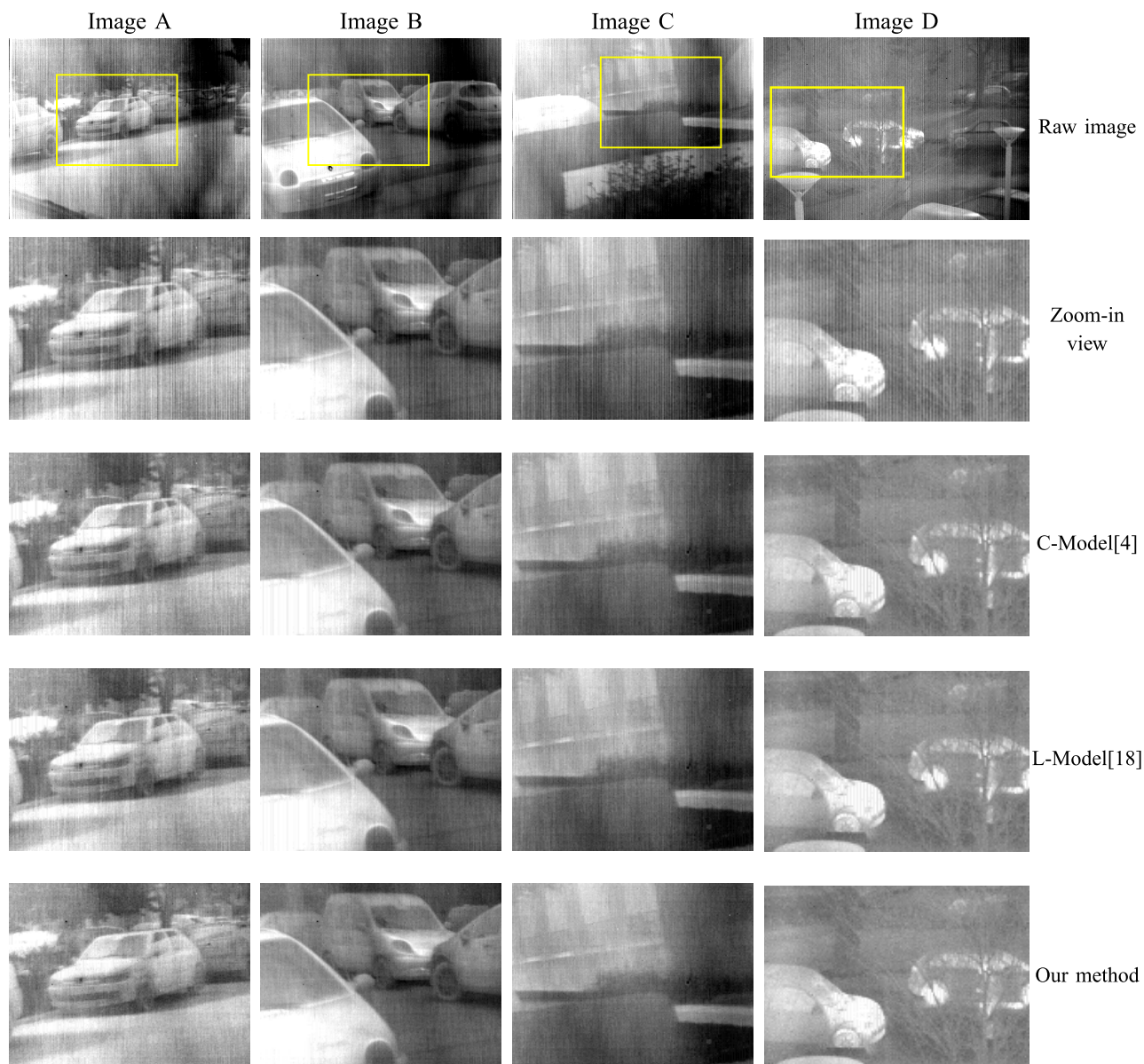


Fig. 13. Comparative results of our proposed method and two model-based strip noise correction methods [4], [18]. Please zoom in to check image details.

adjacent columns is as small as possible. The L-model method proposed in [18] first computes the averages of the first- and second-order statistics for a number of neighboring columns (here we consider nine neighboring columns), and then uses a linear function (gain and offset correction model) to make the pixels from adjacent columns that have similar statistics.

1) *Qualitative Evaluation*: Fig. 13 shows some comparative results of the C-model [4], the L-model [18], and our proposed method. As shown in the zoomed-in patches, the C-model and L-model solutions do not produce satisfactory strip noise reduction results. A significant amount of strips remains visible in their processed results. It is because locally changed strip noise cannot be completely compensated by using column-wise fixed correction factors. In comparison, our proposed algorithm is based on a local correction model, and therefore it can achieve a better performance of strip removal in infrared images as shown in the last row of Fig. 13.

The processing results of the MHE-based method and our proposed method are shown in Fig. 14. It can be observed that our proposed method produces better destriping results than the MHE method. Since the MHE algorithm is based on full-column histogram equalization, it is not capable of removing significant strip noise highlighted in Image E in Fig. 14. As shown in the highlighted region in Image F in Fig. 14, the MHE-based method will generate obvious artifacts in its output. A locally adaptive contrast adjustment algorithm [15], [21] can be implemented to reduce the artifacts, but it involves time-consuming iterations. Another obvious drawback of the MHE-based method is that it will still lose a certain amount of image details. In Fig. 15, we show the estimated strip noise of the zoomed-in patches in Images G and H using different destriping solutions. It is observed that strip noise estimated using the MHE method still contains some image details. This problem will cause performance decrease of

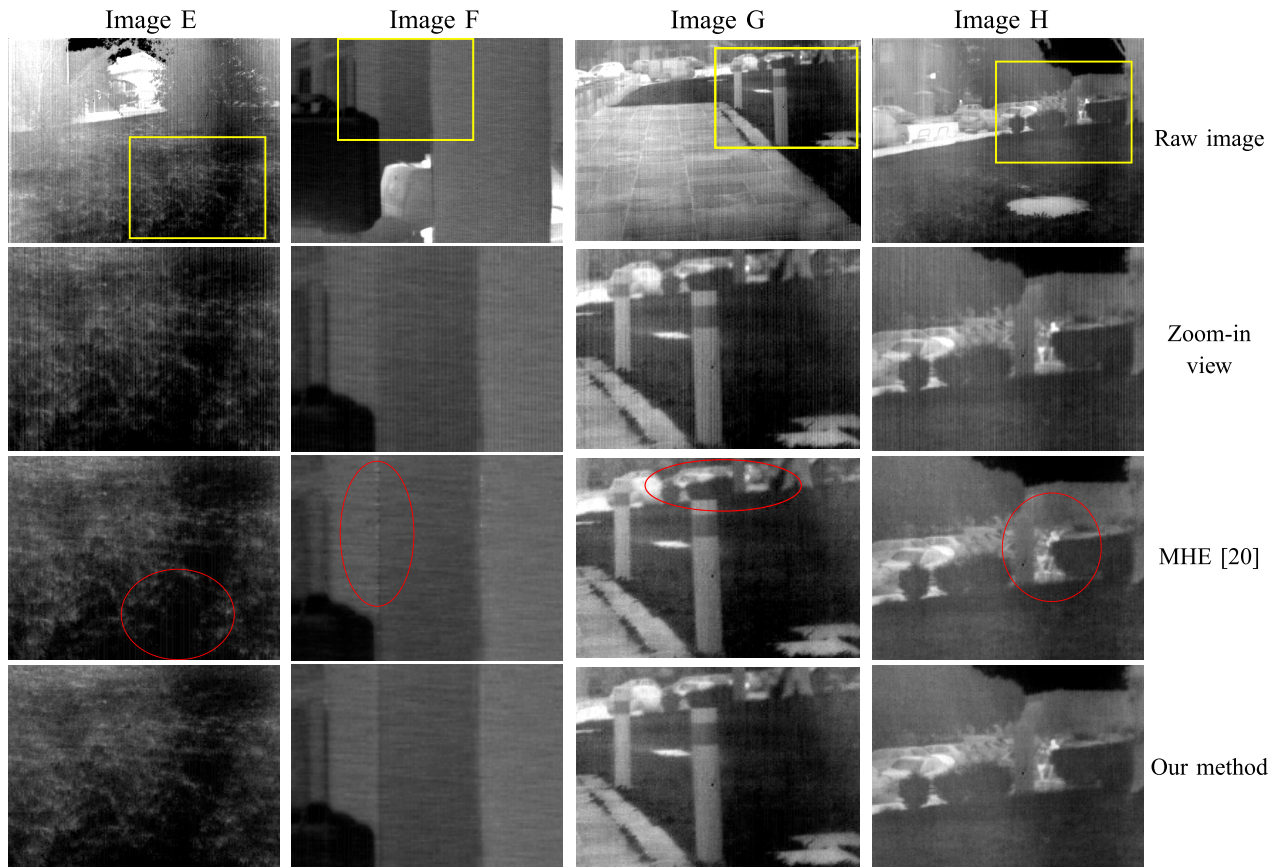


Fig. 14. Comparative results of our proposed method and MHE strip denoising solution [20]. Please zoom in to check the details highlighted in ellipses.

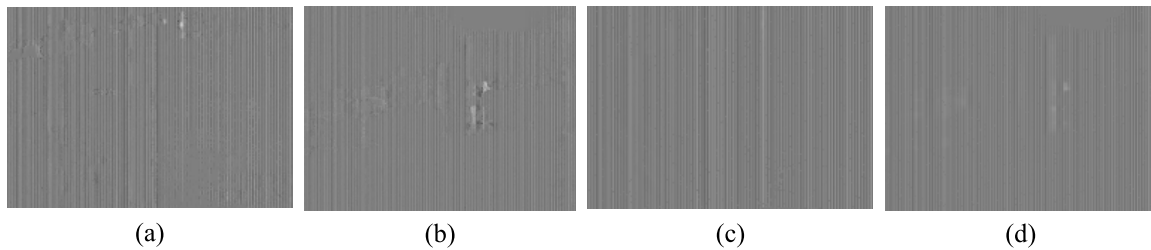


Fig. 15. Estimated strip noise of the zoomed-in patches in Images G and H in Fig. 14. (a) and (b) Strip noise estimated using the MHE method. (c) and (d) Strip noise estimated using our proposed method. The images are normalized to the 0–1 value range by an affine contrast change for visualization.

target detection and recognition for surveillance applications. In comparison, our method can more accurately separate strip noise from image texture. As a consequence, some small targets (e.g., a vehicle behind a tree highlighted in Image G and a pedestrian highlighted in Image H in Fig. 14) are well preserved in our output images, but get lost in the results of the MHE method.

As discussed in Section III-D, our proposed strip noise reduction method can be easily implemented in an FPGA as a preprocessing step to improve the image quality of infrared cameras. It is based on a sequence of 1-D box filters that can be efficiently implemented in an FPGA using a fast calculation algorithm [39]. In comparison, the MHE-based method [21] needs to perform complex column-based histogram equalization and inverse histogram computation, which are too slow for real-time hardware implementation. Moreover, our method requires buffering only part of a raw

image to start processing. After receiving $h = (H/4)$ rows of the raw image data, our algorithm can compute denoised output (the horizontal edge-preserving filtering requires one row of data and the vertical strip noise decomposition requires h rows of data, respectively). In comparison, other strip removal solutions [16], [18], [21] need to compute statistics for a full column of pixels, thus require storing an entire image frame to start computing the corrected output for pixels on the first row. Therefore, our method is less memory demanding.

2) *Quantitative Evaluation*: In this section, we perform quantitative evaluation using infrared images covering a wide range of scenarios as shown in Fig. 16 (e.g., indoor or outdoor). For each scene, we first make use of a commercial uncooled LWIR thermal camera to capture 30 strip-noise-compensated images and average them to obtain a strip-noise-free reference image. Then we send serial command to the

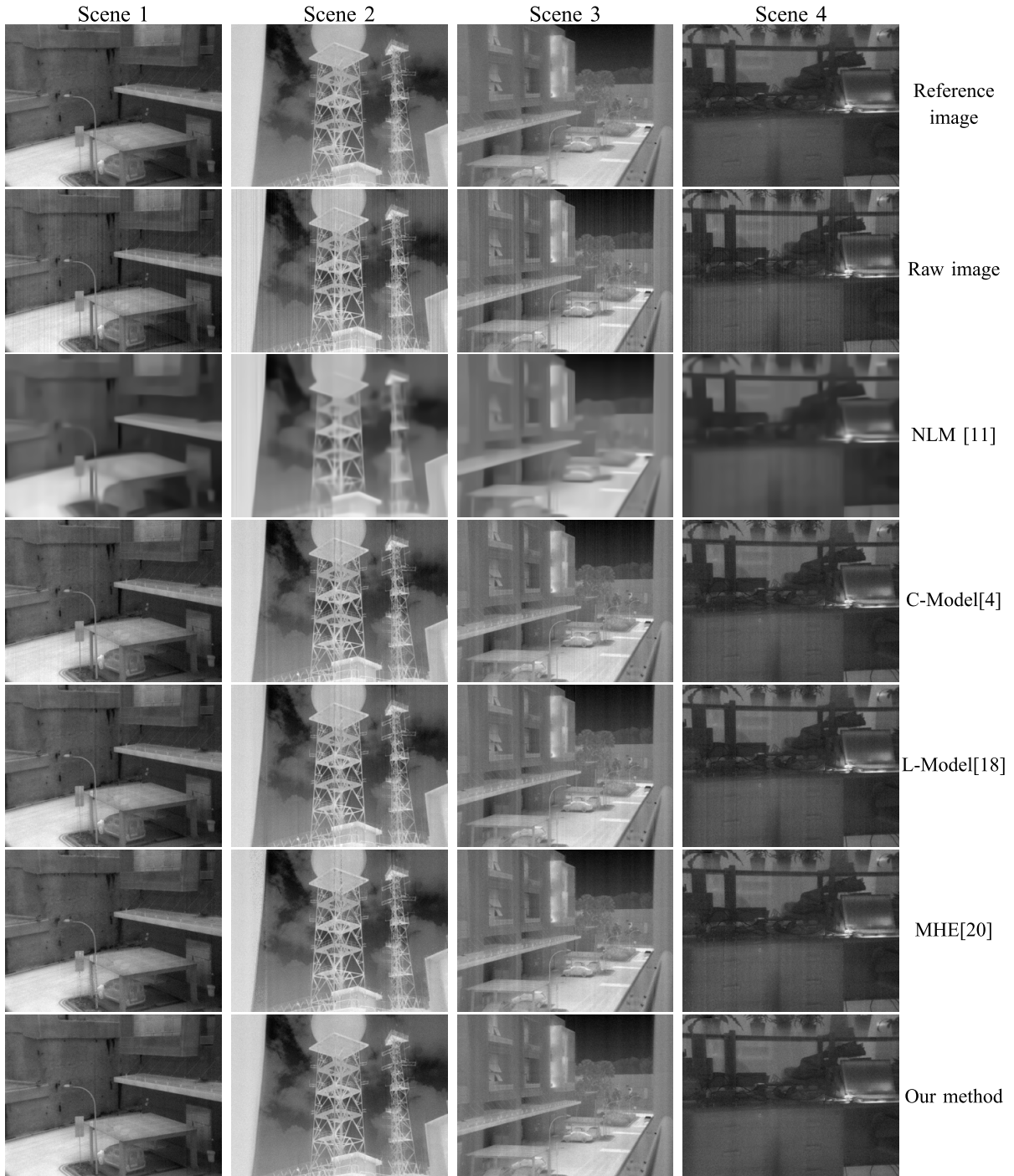


Fig. 16. Comparative results of our proposed method and other state-of-the-art strip denoising solutions [4], [11], [18], [20]. Please zoom in to check details.

camera to turn OFF its on-chip strip noise cancellation algorithm and capture a short infrared image sequence (150 frames for 10 s). Our proposed strip removal algorithm and other 1-D and 2-D [4], [11], [18], [20] denoising methods are individually applied to compute the strip-noise-compensated images. The processing results, partially shown in Fig. 16, confirm that our approach can also achieve the best strip noise removal results for images captured by our own infrared camera. As a quantitative measure of the performance, we compute the Peak signal-to-noise ratio (PSNR) to show the difference between the reference image B and an estimated

image B' as

$$\text{PSNR}(B, B') = 10 \log_{10} \left(\frac{1}{|X|^{-1} \sum_{i \in X} (B_i - B'_i)^2} \right) \quad (11)$$

where X denotes all pixels of an image. Since scene temperature changed during a short period of time (~ 15 s) is negligible, the difference between the reference image and an estimated image provides a good indication of how well different strip removal solutions perform. The quantitative evaluations results are shown in Fig. 17 and Table I.

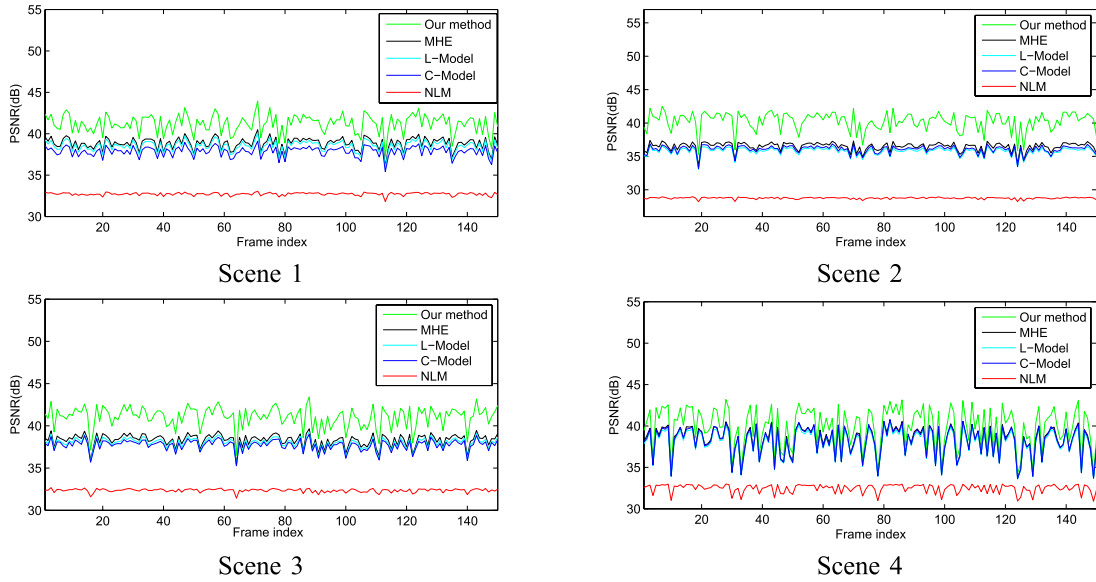


Fig. 17. Estimated PSNR between the reference image and the estimated images using different denoising methods. The difference is often considered as visually insensitive when the PSNR ≥ 40 dB [10].

TABLE I
AVERAGED PSNR BETWEEN THE REFERENCE IMAGE B AND AN
ESTIMATED IMAGE B' PROCESSED USING DIFFERENT
STRIP NOISE CORRECTION SOLUTIONS

	Scene 1	Scene 2	Scene 3	Scene 4
NLM[11]	32.7203	28.7539	32.3577	32.4685
C Model[4]	37.8778	36.0599	37.7424	38.2080
L Model [18]	38.5470	35.8098	37.9963	37.9911
MHE [20]	38.8634	36.5023	38.3493	38.2862
Our method	41.2256	40.2827	41.0688	40.8254

It is observed that the PSNR of our method is significantly higher than the results of other destriping methods. The experimental results illustrate that our proposed method can better eliminate strip noise in infrared images.

V. CONCLUSION

In this paper, we present an effective single-image-based processing algorithm to remove vertical strips in infrared images. It is difficult to remove strip noise without blurring fine details in low-textured infrared images. A two-step approach is proposed for this challenging task. A 1-D row guided filter is first applied to eliminate obvious vertical strip noise. Inevitably, many fine image details are also removed. Then, we apply the second step to retrieve the high-frequency components corresponding to image texture and edges. Our thermal calibration experiment result implies that a local linear relationship exists between the infrared responses and strip noise for sensor detectors within a column. Therefore, the strip noise components are accurately decomposed from the extracted high-frequency signals (contain both strip noise and image texture) by applying a 1-D column guided filter. Only the estimated noise terms are subtracted from the raw infrared images to remove strip noise without blurring image structures. The experimental results show that our proposed method works well with low-textured infrared images and delivers a better performance compared with the state-of-the-art 1-D and 2-D denoising techniques. We plan to implement

the proposed method in our hardware device to improve the quality of infrared images for other high-level computer vision applications, such as video stabilization, stereo matching, image stitching, and target detection and tracking.

REFERENCES

- [1] M. Schulz and L. Caldwell, "Nonuniformity correction and correctability of infrared focal plane arrays," *Infr. Phys. Technol.*, vol. 36, no. 4, pp. 763–777, 1995.
- [2] O. Riou, S. Berrebi, and P. Bremond, "Nonuniformity correction and thermal drift compensation of thermal infrared camera," *Proc. SPIE, Thermosense XXVI*, vol. 5405, pp. 294–302, Apr. 2004.
- [3] Y. Tendero, J. Gilles, S. Landeau, and J. M. Morel, "Efficient single image non-uniformity correction algorithm," *Proc. SPIE, Electro-Opt. Infr. Syst., Technol. Appl. VII*, vol. 7834, p. 78340E, Oct. 2010.
- [4] N. Högastén and M. Ingerhed, "Systems and methods for processing infrared images," U.S. Patent 8 208 026 B2, Jun. 26, 2012.
- [5] D. L. Perry and E. L. Dereniak, "Linear theory of nonuniformity correction in infrared staring sensors," *Opt. Eng.*, vol. 32, no. 8, pp. 1854–1859, 1993.
- [6] J. G. Harris and Y.-M. Chiang, "Nonuniformity correction of infrared image sequences using the constant-statistics constraint," *IEEE Trans. Image Process.*, vol. 8, no. 8, pp. 1148–1151, Aug. 1999.
- [7] R. C. Hardie, M. M. Hayat, E. Armstrong, and B. Yasuda, "Scene-based nonuniformity correction with video sequences and registration," *Appl. Opt.*, vol. 39, no. 8, pp. 1241–1250, 2000.
- [8] A. Averbuch, G. Liron, and B. Z. Bobrovsky, "Scene based non-uniformity correction in thermal images using Kalman filter," *Image Vis. Comput.*, vol. 25, no. 6, pp. 833–851, 2007.
- [9] D. R. Pipa, E. A. B. da Silva, C. L. Pagliari, and P. S. R. Diniz, "Recursive algorithms for bias and gain nonuniformity correction in infrared videos," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4758–4769, Dec. 2012.
- [10] K. He, J. Sun, and X. Tang, "Guided image filtering," in *Computer Vision*. Berlin, Germany: Springer-Verlag, 2010, pp. 1–14.
- [11] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, Jun. 2005, pp. 60–65.
- [12] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. 6th Int. Conf. Comput. Vis.*, Jan. 1998, pp. 839–846.
- [13] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.
- [14] N. J. W. Morris, S. Avidan, W. Matusik, and H. Pfister, "Statistics of infrared images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2007, pp. 1–7.

- [15] X. Sui, Q. Chen, and G. Gu, "Adaptive grayscale adjustment-based stripe noise removal method of single image," *Infr. Phys. Technol.*, vol. 60, pp. 121–128, Sep. 2013.
- [16] W. Qian, Q. Chen, G. Gu, and Z. Guan, "Correction method for stripe nonuniformity," *Appl. Opt.*, vol. 49, no. 10, pp. 1764–1773, 2010.
- [17] J. Zhao *et al.*, "Single image stripe nonuniformity correction with gradient-constrained optimization model for infrared focal plane arrays," *Opt. Commun.*, vol. 296, pp. 47–52, Jun. 2013.
- [18] B. Narayanan, R. C. Hardie, and R. A. Muse, "Scene-based nonuniformity correction technique that exploits knowledge of the focal-plane array readout architecture," *Appl. Opt.*, vol. 44, no. 17, pp. 3482–3491, 2005.
- [19] Y. Cao and Y. Li, "Strip non-uniformity correction in uncooled long-wave infrared focal plane array based on noise source characterization," *Opt. Commun.*, vol. 339, pp. 236–242, Mar. 2015.
- [20] Y. Tenderso, S. Landeau, and J. Gilles, "Non-uniformity correction of infrared images by midway equalization," *Image Process. Line*, vol. 2, pp. 134–146, Jul. 2012. [Online]. Available: http://demo.ipol.im/demo/glm_t_mire/
- [21] Y. Tenderso and J. Gilles, "ADMIRE: A locally adaptive single-image, non-uniformity correction and denoising algorithm: Application to uncooled IR camera," *Proc. SPIE, Infr. Technol. Appl. XXXVIII*, vol. 8353, p. 83531O, May 2012.
- [22] J. Delon, "Midway image equalization," *J. Math. Imag. Vis.*, vol. 21, no. 2, pp. 119–134, 2004.
- [23] L. Yuan, J. Sun, L. Quan, and H.-Y. Shum, "Progressive inter-scale and intra-scale non-blind image deconvolution," *ACM Trans. Graph.*, vol. 27, no. 3, 2008, Art. ID 74.
- [24] J. M. Mooney, F. D. Sheppard, W. S. Ewing, J. E. Ewing, and J. Silverman, "Responsivity nonuniformity limited performance of infrared staring cameras," *Opt. Eng.*, vol. 28, no. 11, p. 281151, 1989.
- [25] Y. Cao and C.-L. Tisse, "Single-image-based solution for optics temperature-dependent nonuniformity correction in an uncooled long-wave infrared camera," *Opt. Lett.*, vol. 39, no. 3, pp. 646–648, 2014.
- [26] G. Bieszczad, T. Orzanowski, T. Sosnowski, and M. Kasteck, "Method of detectors offset correction in thermovision camera with uncooled microbolometric focal plane array," *Proc. SPIE, Electro-Opt. Infr. Syst., Technol. Appl. VI*, vol. 7481, pp. 74810O-1–74810O-8, Sep. 2009.
- [27] E. Vera, P. Meza, and S. Torres, "Total variation approach for adaptive nonuniformity correction in focal-plane arrays," *Opt. Lett.*, vol. 36, no. 2, pp. 172–174, 2011.
- [28] L. Shang and D. Huang, "Image denoising using non-negative sparse coding shrinkage algorithm," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 1017–1022.
- [29] G. Y. Chen, T. D. Bui, and A. Krzyzak, "Image denoising with neighbour dependency and customized wavelet and threshold," *Pattern Recognit.*, vol. 38, no. 1, pp. 115–124, 2005.
- [30] Y. Sanada *et al.*, "FPGA implementation of a memory-efficient stereo vision algorithm based on 1-D guided filtering," in *Proc. Int. Conf. Circuits, Syst., Control*, 2014, pp. 25–30.
- [31] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.
- [32] Y. Ding, J. Xiao, and J. Yu, "Importance filtering for image retargeting," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 89–96.
- [33] A. Y.-S. Chia *et al.*, "Semantic colorization with Internet images," *ACM Trans. Graph.*, vol. 30, no. 6, 2011, Art. ID 156.
- [34] P. Bauszat, M. Eisemann, and M. Magnor, "Guided image filtering for interactive high-quality global illumination," *Comput. Graph. Forum*, vol. 30, no. 4, pp. 1361–1368, 2011.
- [35] C. Rhemann, A. Hosni, M. Bleyer, C. Rother, and M. Gelautz, "Fast cost-volume filtering for visual correspondence and beyond," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 3017–3024.
- [36] Q. Zhang, X. Shen, L. Xu, and J. Jia, "Rolling guidance filter," in *Computer Vision*. New York, NY, USA: Springer-Verlag, 2014, pp. 815–830.
- [37] A. Levin and B. Nadler, "Natural image denoising: Optimality and inherent bounds," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 2833–2840.
- [38] K. Ohata *et al.*, "Hardware-oriented stereo vision algorithm based on 1-D guided filtering and its FPGA implementation," in *Proc. IEEE 20th Int. Conf. Electron., Circuits, Syst. (ICECS)*, Dec. 2013, pp. 169–172.
- [39] M. J. McDonnell, "Box-filtering techniques," *Comput. Graph. Image Process.*, vol. 17, no. 1, pp. 65–70, 1981.



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